

FATIMA MAMU

AI Product Engineer | Voice AI & Real-Time Systems | LLM Applications

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PROFESSIONAL SUMMARY

AI Product Engineer specializing in production-grade voice AI systems, real-time speech pipelines, and LLM-powered applications. Hands-on builder with end-to-end ownership across discovery, architecture, prototyping, deployment, and customer-facing iteration. Combines strong software engineering fundamentals (Python, TypeScript/JavaScript) with deep, applied experience in speech-to-text (STT), text-to-speech (TTS), real-time audio streaming, retrieval-augmented generation (RAG), and agentic workflows. Proven track record of translating ambiguous customer problems into shipped products — including voice agents deployed across clinical environments, embedded directly with end users to validate behavior, optimize latency, and earn the technical win. Comfortable as the technical authority bridging engineering and customers: leading discovery, scoping POCs, building lightweight prototypes with SDKs and APIs, and communicating complex AI/ML concepts to both technical and non-technical stakeholders.

TECHNICAL SKILLS

Languages: Python, TypeScript / JavaScript, SQL; working knowledge of Rust

Voice & Real-Time AI: Speech-to-Text (STT), Text-to-Speech (TTS), real-time streaming audio, low-latency voice agents, conversational AI, multi-turn dialog, telephony integrations, WebSockets, WebRTC, voice activity detection

LLMs & AI Systems: LLM application development, retrieval-augmented generation (RAG), agentic and multi-step workflows, prompt engineering, function calling / tool use, evaluation harnesses and benchmarking, structured outputs, human-in-the-loop pipelines, fine-tuning fundamentals (post-training)

APIs & Developer Tools: FastAPI, REST and streaming APIs, Gemini Live, Gemini 2.5 Flash, OpenAI / Anthropic APIs, JotForm, webhook-driven integrations, SDK-based prototyping

Cloud & Infrastructure: Google Cloud (Cloud Run, Cloud SQL, Secret Manager), AWS fundamentals, PostgreSQL, Supabase, pgvector (vector search), Docker, containerized deployments, CI/CD

Customer-Facing Engineering: Technical discovery, proof-of-concept (POC) design and execution, live demos, architecture reviews, latency / accuracy / cost trade-off analysis, on-site enterprise customer engagement, requirements translation, technical writing

Computer Vision & Perception: 6-DoF pose estimation, synthetic data generation, real-time motion tracking, AR/spatial systems, Unity, TouchDesigner

PROFESSIONAL EXPERIENCE

Founding Engineer | The One Care, Boston, MA

2026 – Present

- Architected and shipped a production voice-based caregiving AI agent built on a real-time STT → reasoning → TTS pipeline, combining structured retrieval over pgvector / PostgreSQL with an LLM reasoning layer to deliver multi-turn, context-aware interactions; achieved ~25% improvement in response relevance across ~1,500 evaluation queries.
- Designed and built an end-to-end voice-to-structured-data pipeline that converts unstructured caregiver speech into validated clinical records in real time, with schema-level validation, latency budgets tuned for live interaction, and graceful handling of disfluent, accented, and incomplete inputs.
- Embedded directly with caregivers as the primary technical contact — running discovery sessions, observing live usage, triaging issues, and translating field signals into prioritized system improvements; functioned as the bridge between end users and engineering.
- Built an automated ingestion and classification pipeline using Gemini 2.5 Flash with a human-in-the-loop review layer to scale curation of caregiver experiences into a high-quality, retrieval-ready knowledge index.

- Stood up the full cloud footprint on Google Cloud (Cloud Run, Cloud SQL, Secret Manager) and built a FastAPI-based evaluation harness with structured test cases covering ambiguous inputs, incomplete context, and conflicting caregiver reports — used as the gate before any production deploy.
- Owned cost / latency / accuracy trade-offs across the stack, instrumenting the pipeline with structured logging and metrics so model and prompt changes could be evaluated against real production traffic, not just offline benchmarks.

AI Product Engineer | Lexi AI, Boston, MA

2025 – 2026

- Shipped customer-facing real-time voice intelligence features deployed across 4 clinical facilities and 7,000+ live conversations, improving clinician efficiency by ~40% through streaming audio understanding, real-time visual cues, and workflow-integrated voice controls.
- Designed, built, and deployed a phone-based AI intake voice agent on Gemini Live, handling end-to-end real-time speech: low-latency STT, conversational reasoning, structured data extraction, and API-based routing into provider systems (JotForm) — built specifically for low-digital-literacy users on standard PSTN telephony.
- Operated as the on-site technical lead in live clinical environments: shadowing clinicians, diagnosing voice-system breakdowns in real workflows, and turning observations into reliability, UX, and prompt improvements that were shipped within the same sprint.
- Owned the full pre-sales-style technical narrative for prospective facilities — running live demos, answering deep technical questions on accuracy, latency, integration patterns, and data handling, and producing clear written materials for technical and non-technical stakeholders.
- Built internal evaluation tooling to track real-world transcription quality, agent task-completion rates, and end-to-end conversation latency, surfacing regressions before they affected customers.

Software Development Intern, R&D | Walt Disney Imagineering, Glendale, CA

2024

- Built internal tools for the R&D team, including a synthetic dataset generation pipeline for 6-DoF pose estimation and an AR pre-production staging environment, reducing physical setup time and improving spatial accuracy before on-site installation.
- Developed real-time, sensor-driven interactive experiences in Unity, Maya, and TouchDesigner — including a guest-facing water simulation for the Moana: Journey of Water exhibit (D23) that translated full-body motion tracking into responsive visual effects with sub-frame latency requirements.
- Integrated robotics, real-time animation, and physical staging for a KUKA robot-based illusion, synchronizing TouchDesigner animations with LED outputs; presented the working system to Bob Iger, securing approval for project continuation.
- Worked across hardware, software, and creative stakeholders — practicing the same translation skill required in customer-facing technical roles: making complex real-time systems legible to non-engineers.

Engineering Associate (Prototyper) | The Business People Limited, Nigeria

2019 – 2023

- Acted as the primary technical lead on client engagements across agriculture, aviation, and healthcare — owning requirements discovery, system design, stakeholder communication, and coordination with external engineering teams, equivalent to a hands-on solutions engineer / pre-sales architect role.
- Designed and built full-stack prototypes (Next.js, TypeScript, PostgreSQL) covering data modeling, API design, and end-to-end user flows; used these prototypes to validate product direction with stakeholders and de-risk full-scale builds before commit.
- Translated ambiguous operational challenges from non-technical clients into concrete technical specifications, evaluation criteria, and demo-ready POCs — directly analogous to leading technical discovery and POC design in an enterprise sales cycle.
- Presented technical solutions to executive and operational stakeholders, defending architecture choices on cost, scalability, and reliability, and iterating based on feedback from real users.

Founder & Creative Director | Edible Art Studio, Nigeria

2013 – 2023

- Founded and ran a bespoke design studio with a 15-person team, delivering thousands of commissioned client projects over a decade — built business and operational instincts that translate directly into product thinking and customer empathy.

- Designed and implemented operational systems for client intake, custom order management, and supplier coordination, enabling consistent, scalable delivery and laying the foundation for a later transition into engineering and AI product roles.

EDUCATION

Harvard University, Cambridge, MA, USA — Master in Design Engineering (MDE) 2023 – 2025

MDE Fellowship recipient (merit-based); Leadership Award for cross-disciplinary collaboration.

University College London (UCL), London, UK — Diploma in Computer Graphics, Vision and Imaging 2015 – 2016

Brunel University London, London, UK — BEng, Electrical and Electronics Engineering 2010 – 2013

HONORS, AWARDS & CERTIFICATIONS

- **Harvard University** — Leadership Award (2025); MDE Fellowship (2024).
- **Microsoft** Full-Stack Developer Professional Certificate.
- **IBM** AI Product Manager Professional Certificate.
- **Specialized coursework:** Machine Learning in Production; Fine-Tuning & Reinforcement Learning for LLMs (Intro to Post-Training); A2A — The Agent2Agent Protocol; Leveraging Data in Iterative Product Design.

SELECTED PROJECTS

- **Voice Caregiving Agent (Production)** — Real-time STT/TTS voice assistant with pgvector RAG and LLM reasoning over caregiver knowledge; demonstrates end-to-end ownership of a low-latency voice AI product.
- **Phone-Based AI Intake Agent** — Telephony-integrated voice agent on Gemini Live for low-digital-literacy users, with structured data extraction and routing to downstream provider systems.
- **Evaluation Harness for Voice Agents** — FastAPI-based test framework with curated edge-case datasets (ambiguous, incomplete, conflicting inputs), used to gate model and prompt changes pre-deployment.
- **Real-Time Interactive AR / Motion-Tracking Experiences** — Unity + TouchDesigner systems built at Disney Imagineering R&D, including the Moana: Journey of Water D23 exhibit.

WHAT I BRING TO A VOICE AI SOLUTIONS TEAM

- Built and shipped real, production voice agents end-to-end — not just prototypes — including streaming STT pipelines, LLM reasoning layers, and structured output extraction running in live clinical environments.
- Customer-facing instinct: comfortable embedding with users, running discovery, demoing live, defending architectural trade-offs (latency, accuracy, cost, deployment model) to both engineers and executives.
- AI-first operating style: heavy daily use of LLMs, agents, and AI coding tools; consistently integrates new models (Gemini Live, Gemini 2.5 Flash, etc.) into workflows the day they ship.
- Translator of ambiguity: a decade of running client-facing engagements before engineering — equally fluent talking to a CTO, a clinician, or a non-technical end user about the same system.